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Education

NIT Surat	B.Tech - Artificial Intelligence	7.4 CGPA	2023-Present
SGVP, Rajkot	12th (CBSE)	87.1%	2023
Kendriya Vidyalaya, Rajkot	10th (CBSE)	91.2%	2021

Projects

Research

Can You Break RLVER? Probing Adversarial Robustness of RL-Trained Empathetic Agents

Reinforcement Learning, LLM Evaluation, Empathy Modeling

- Designed the **Adversarial Empathy Benchmark (AEB)** with six psychologically grounded adversarial dialogue trajectories including *gaslighting, emotional escalation, and manipulation for validation.*
- Proposed **Emotional Consistency Score (ECS)**, a confidence-weighted metric to distinguish between **emotional state tracking** and **state improvement** in dialogue systems.
- Conducted a **controlled scenario-matched experiment** across **480 adversarial dialogues** evaluating RL-trained (PPO, GRPO) and baseline LLMs under identical conditions.
- Demonstrated that **RLVER training improves emotional responsiveness** (Final Score: 0.963 vs 0.761 baseline, $p < 0.001$) while **not improving state tracking**, establishing the ECS–FS dissociation.
- Identified **Thinking Scaffold Reversal**: chain-of-thought reasoning degrades untuned models but significantly improves RL-trained agents.
- Showed **+47% improvement in hidden intention detection** and **zero dialogue collapse** for RL-trained models under adversarial conditions.
- **Manuscript under review at NeurIPS 2026.**

VidSynth AI – YouTube Chat Assistant

Live Demo

- Developed an intelligent **Streamlit application** that transforms YouTube videos into structured notes, key topics, and an interactive chatbot interface.
- Implemented a **Retrieval-Augmented Generation (RAG)** system using **Gemini 2.5 Flash Lite** and **ChromaDB** for context-aware video-based question answering.
- Engineered modules for transcript extraction, English translation, topic detection, and concise note generation using **LangChain** and **Sentence Transformers**.
- Designed and deployed a user-friendly web app on **Huggingface**, enabling real-time video interaction and content summarization.

Rare Gen master - Synthetic Data Generation for Rare Cancer Diseases using GANs

Deep Learning, PyTorch

- Building a **GAN-based framework** (WGAN-GP, DCGAN) to synthesize rare cancer imaging data, addressing dataset scarcity and imbalance.
- Validating generated data with **FID/IS metrics** and testing improvements in downstream **classification performance**.
- Ensuring **privacy-preserving synthetic data** for safe use in healthcare ML applications.

Skills

Programming Languages: C, C++, Python
Web Development: HTML
Databases: MongoDB, MySQL,Cassandra
Artificial Intelligence & Machine Learning: Machine Learning, Deep Learning, Reinforcement Learning, Supervised & Unsupervised Learning, Neural Networks, NLP (Basics)
AI/ML Libraries: Scikit-learn, TensorFlow, Keras, PyTorch, Hugging Face,Transformers
Data Science & Visualization: Pandas, NumPy, Matplotlib, Seaborn , R Language
Gen AI frameworks: Langchain
Tools & Platforms: Git,Colab,Kaggle, GitHub, Jupyter Notebook, Anaconda, VS Code

Positions of Responsibility

- **Manager**, *MINDBEND (Annual Technical Festival)* — **Jan 2026 – Apr 2026**
- **Co-Head**, *MINDBEND (Annual Technical Festival)* — **Jan 2025 – Apr 2025**
- **Executive**, *MINDBEND (Annual Technical Festival)* — **Jan 2024 – Apr 2024**